M5T1 BACHY105 · 8 Hours · Quantum Methods, Basis Sets, LCAO–MO, PEC vs. PES, MEP, SN2 Profiles & AI Screening

Computational Chemistry & AI Material Discovery

Complete editorial study notes covering Born–Oppenheimer approximation, HF, DFT, MP2, CCSD(T) & CASSCF methods, Pople & Dunning basis sets, homonuclear (H2) vs. heteronuclear (NO) MO diagrams, 1D PEC vs. multi-D PES, NH3 electrostatic potential maps, vibrational modes, gas vs. aqueous SN2 profiles, and AI semiconductor discovery.

○ Mark as done ◜ Highlight Quiz this topic

All syllabus sections included · Ticks itself when every tab is read

Definitions ✓ Formulas Key Concepts Diagrams & Derivations Solved Examples

M5S0 · MEMORY STRIP

Quick-Reference Equations, Degrees of Freedom & Frequency Rules

Time-Independent Schrödinger Eq.	$\hat{H}\,\Psi(r;R) = E\,\Psi(r;R)$ — Born–Oppenheimer: $M_{nucleus} \gg m_e$ (≥1836×) decouples nuclear & electronic motion
LCAO–MO & Bond Order	$\Psi_{\pm} = c_A\psi_A \pm c_B\psi_B$ | $BO = \frac{1}{2}(N_b - N_a)$ — H_2: BO = 1.0 (diamagnetic) | NO (11 valence e^-): BO = ½(8−3) = 2.5 (paramagnetic)
Internal / Vibrational Modes	**Non-Linear: 3N − 6** | **Linear: 3N − 5** — H_2O (N=3): 3 modes · CO_2 (N=3, linear): 4 modes · NH_3 (N=4): 6 modes · CH_4 (N=5): 9 modes
Stationary Point Hessian Rule	$\tilde{\nu} = (1 / 2\pi c)\sqrt{(k / \mu)}$ — Minimum (Reactant/Intermediate): **0 imaginary freq.** · Transition State: **1 imaginary freq.**

M5S1 · 7 QM METHODS + 6 BASIS SET FAMILIES

Computational Chemistry Hierarchy, Electronic Structure Methods & Basis Sets

Classification & 7 Merits of Computational Chemistry
QM vs. Classical MM/MD

Two Major Branches: (1) **Quantum Mechanics (Electronic Structure):** Solves Schrödinger's equation explicitly for electrons (*Ab-Initio, DFT, Semi-Empirical*) — capable of modeling **bond breaking/forming, transition states, and excited states**; (2) **Classical Mechanics (Force-Field):** Treats atoms as balls and bonds as Newtonian springs (*Molecular Mechanics MM & Molecular Dynamics MD*) for large proteins/polymers (>10^5 atoms); (3) **Hybrid QM/MM:** Treats the reactive enzyme active site with QM and surrounding protein/solvent with MM.

7 Merits Over Experiment: (1) Characterizes fleeting **transition states (TS, lifetime ~10^{-13} s)** and unstable intermediates; (2) Zero chemical waste or exposure to toxic/explosive/radioactive species; (3) Accesses extreme pressures/temperatures (planetary cores, combustion); (4) Visualizes 3D MOs, spin densities, and MEP maps; (5) Decomposes steric vs. electronic energy terms; (6) Screens thousands of drug/battery/semiconductor candidates before synthesis; (7) Interprets ambiguous experimental spectra (IR, NMR, UV–Vis).

TDSE vs. TISE & Born–Oppenheimer Approximation Foundational Postulate	**Time-Dependent (TDSE, $i\hbar\,\partial\Psi/\partial t = \hat{H}\Psi$):** Used for ultrafast laser dynamics and wavepacket propagation. **Time-Independent (TISE, $\hat{H}\Psi = E\Psi$):** Used for stationary molecular states, geometry optimization, and PES mapping, where $\hat{H} = \hat{T}_n + \hat{T}_e + \hat{V}_{ne} + \hat{V}_{ee} + \hat{V}_{nn}$. **Born–Oppenheimer Approximation:** Since nuclei are at least **1836× heavier than electrons**, electrons adjust instantaneously to any nuclear movement. Setting nuclear kinetic energy $\hat{T}_n = 0$ allows solving the electronic Schrödinger equation at fixed nuclear coordinates **R**, generating the **Potential Energy Surface $E(R) = E_{elec}(R) + V_{nn}(R)$**.

Electronic Structure Method	Physical Approximation & Electron Correlation	Scaling & Accuracy	Best Use Cases
1. Hartree–Fock (HF)	Single Slater determinant; each e^- moves in the **mean field** of all other e^-; **neglects dynamic Coulomb correlation**	~N^4 · Low accuracy for reaction barriers	Initial geometry guesses & baseline reference wavefunction
2. Density Functional Theory (DFT: B3LYP, M06-2X)	Replaces 3N-variable wavefunction with 3D **electron density ρ(r)** + Exchange-Correlation functional $E_{xc}[\rho]$	~N^3 · High accuracy-to-cost ratio (Workhorse!)	Geometry, frequencies, reaction mechanisms (50–500 atoms), transition-metal catalysis
3. MP2 (Møller–Plesset 2nd Order)	Adds 2nd-order Rayleigh–Schrödinger perturbation to HF to recover ~85–90% dynamic correlation	~N^5 · Good for non-covalent interactions	Hydrogen bonding, π–π stacking, conformational energies
4. CI (Configuration Interaction)	Linear combination of ground + excited Slater determinants (CISD / Full CI)	~N^6 to N! · Variational	Excited states and benchmark small-molecule wavefunctions
5. CCSD(T) (Coupled Cluster)	Exponential cluster operator e^{T} with singles, doubles & perturbative triples	~N^7 · **"Gold Standard" (±1 kcal/mol)**	High-precision thermochemistry and benchmark barrier heights (<30 atoms)
6. MCSCF & 7. CASSCF	Optimizes both CI expansion coefficients and MO orbitals across a Complete Active Space (CAS)	Exponential with active space	**Multi-reference systems:** homolytic bond breaking, diradicals, photochemical conical intersections

Basis Set Family	Notation Example	Mathematical Description & Purpose
1. Minimal Basis Set	**STO-3G**	Exactly 1 basis function per core and valence atomic orbital (3 primitive Gaussians fitted to 1 Slater orbital); fast but inflexible.
2. Split-Valence (Pople)	**6-31G, 6-311G**	Core orbitals represented by 1 contracted block (6 PGTOs); valence orbitals split into inner (3 PGTOs) and outer (1 PGTO) functions so bonds can expand/contract.
3. Polarization Functions (* or d,p)	**6-31G(d,p) / 6-31G****	Adds higher angular-momentum orbitals (**d-orbitals on C/N/O; p-orbitals on H**) to allow asymmetric angular distortion of electron clouds during bonding.

Basis Set Family	Notation Example	Mathematical Description & Purpose
4. Diffuse Functions (+ or ++)	**6-31+G(d,p)**	Adds spatially large, small-exponent s and p functions essential for **anions (Cl^-), lone pairs, hydrogen bonds, and Rydberg states**.
5. Correlation-Consistent (Dunning)	**cc-pVDZ, cc-pVTZ, aug-cc-pVTZ**	Systematically converges post-HF correlation energy toward the Complete Basis Set (CBS) limit (**aug-** adds diffuse functions).
6. Effective Core Potential (ECP)	**LANL2DZ, Def2-TZVP**	Replaces inner core electrons of heavy transition metals (Ru, Pt, Au) with a pseudopotential that accounts for **relativistic effects** while treating valence electrons explicitly.

M5S2 · 3 LCAO RULES + HOMO/LUMO + H_2 & NO LADDERS

Atomic Orbitals, LCAO–MO Theory, Homonuclear (H_2) vs. Heteronuclear (NO) MO Diagrams

Quantum Numbers & 3 LCAO Criteria

$\Psi = c_A\psi_A \pm c_B\psi_B$

4 Quantum Numbers & Filling Rules: Principal n (size/energy), Azimuthal l = 0..n−1 (shape: s, p, d, f), Magnetic m_l = −l..+l (orientation), Spin m_s = ±½; governed by **Aufbau Principle**, **Pauli Exclusion Principle**, and **Hund's Rule of Maximum Multiplicity**.

3 Mandatory Conditions for LCAO Combination: (1) **Comparable Energy** between combining atomic orbitals ψ_A and ψ_B; (2) **Significant Spatial Overlap** along the internuclear axis; (3) **Identical Symmetry** with respect to the internuclear axis (e.g., s + p_z → σ; p_x + p_x → π; s + p_x gives zero net overlap).

Bonding (Ψ_b) vs. Antibonding (Ψ^*): Constructive in-phase addition concentrates electron density between nuclei, lowering energy (**$E_b < E_{AO}$**); destructive out-of-phase subtraction creates a **nodal plane** between nuclei, raising energy (**$E^* > E_{AO}$**; antibonding destabilization ΔE^* exceeds bonding stabilization ΔE_b).

Homonuclear (H_2) vs. Heteronuclear (NO) MO Diagrams Official Slide & Q3 Comparison	**1. Homonuclear Diatomic H_2 (2 Electrons, Equal AO Energies):** Configuration: **$(\sigma_{1s})^2\ (\sigma^*_{1s})^0$** | Equal LCAO coefficients ($c_A = c_B$, non-polar symmetric MOs). **Bond Order:** BO = ½(2 − 0) = **1.0** (R_e = 0.74 Å, D_e = 458 kJ mol^{-1}); **Diamagnetic** (all electrons paired); **HOMO = σ_{1s}, LUMO = σ^*_{1s}**. **2. Heteronuclear Diatomic Nitric Oxide NO (11 Valence Electrons: N = $2s^2 2p^3$ [5], O = $2s^2 2p^4$ [6]):** **Asymmetric AO Energy Levels:** Because Oxygen is more electronegative than Nitrogen (χ_O = 3.44 > χ_N = 3.04), **Oxygen's 2s and 2p atomic orbitals lie at lower energy than Nitrogen's 2s and 2p orbitals**! **Orbital Polarization:** Bonding MOs (σ_{2s}, π_{2p}, σ_{2p}) have larger coefficients on **Oxygen ($c_O > c_N$)**, whereas antibonding MOs (π^*_{2p}, σ^*_{2p}) are localized predominantly on **Nitrogen ($c_N > c_O$)**. **Valence MO Configuration (11 e$^-$): $(\sigma_{2s})^2\ (\sigma^*_{2s})^2\ (\pi_{2px})^2\ (\pi_{2py})^2\ (\sigma_{2pz})^2\ (\pi^*_{2px})^1$**. **Bond Order & Magnetism: BO = ½($N_b - N_a$) = ½(8 − 3) = 2.5**; **Paramagnetic radical** (1 unpaired electron in **π^*_{2p} SOMO/HOMO** localized mainly on N). *Exam Tip:* Ionizing NO → NO^+ removes the antibonding π^*_{2p} electron, increasing BO from 2.5 → **3.0** and shortening the bond!

M5S3 · H_2/HCL PEC + OZONE & H_2O PES

Potential Energy Curves (PEC) vs. Potential Energy Surfaces (PES: Ozone & Water)

Feature	Potential Energy Curve (1D PEC)	Potential Energy Surface (Multi-D PES / Hypersurface)
System & Dimensionality	**Diatomic molecules only (N = 2):** Energy E plotted against a single internuclear distance *r* (2D graph)	**Polyatomic molecules or reactions (N ≥ 3):** Energy plotted over **3N − 6** (or 3N − 5) internal coordinates (3D surface for 2 coordinates; **Hypersurface** when coordinates ≥ 3)
Stationary Points	Single energy well minimum at equilibrium bond length **r_e** (depth = dissociation energy D_e; zero-point corrected = D_0)	Multiple **Local/Global Minima** (reactants, products, intermediates) connected by **First-Order Saddle Points (Transition States)**
Slide Examples	**H_2 PEC** (r_e = 0.74 Å, D_e = 4.75 eV) & **HCl PEC** (r_e = 1.27 Å, D_e = 4.62 eV)	**Ozone (O_3) isomerization PES** (open C_{2v} 120.9° vs. cyclic D_{3h} 60° iso-ozone) & **Water (H_2O) bending/stretching PES** (r_{OH} = 0.958 Å, θ_{HOH} = 104.5°)

Harmonic vs. Anharmonic Morse PEC
Why Bonds Dissociate

Harmonic Approximation: Parabolic curve **$V(r) = \frac{1}{2} k (r - r_e)^2$**; valid only near the bottom of the well ($r \approx r_e$) and fails to predict bond dissociation at large *r*.

Anharmonic Morse Potential (Real Molecule): $V(r) = D_e [1 - \exp(-a(r - r_e))]^2$. At short $r < r_e$, potential rises steeply due to V_{nn} nuclear repulsion; at $r = r_e$, net force is zero (global minimum); as $r \to \infty$, the curve flattens asymptotically to the bond dissociation limit (homolytic cleavage into neutral radicals in gas phase). Note: **$D_e = D_o + ZPE$**, where Zero-Point Energy ZPE = $\frac{1}{2}h\nu_o$.

PES Stationary Points & Ozone / Water Slide Cases
Minima vs. Saddle Points

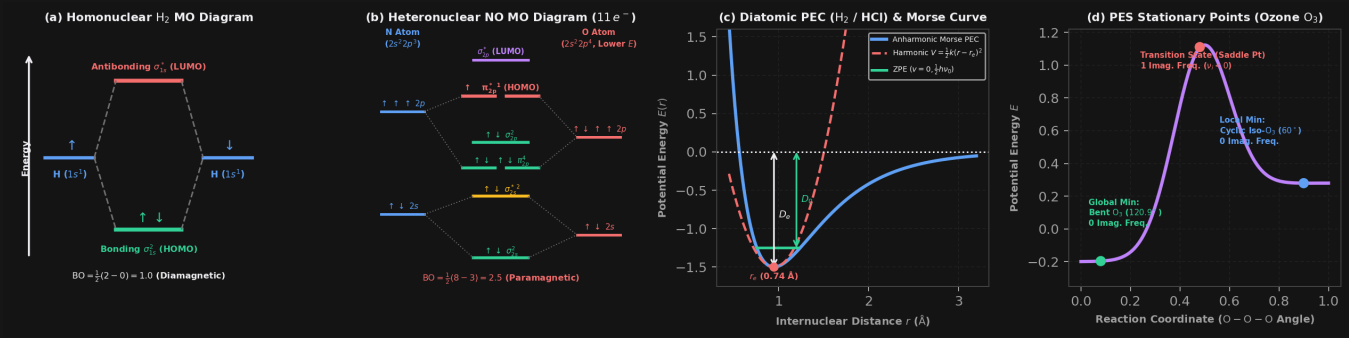
Mathematical Criteria on a PES (Gradient $g = \nabla E = 0$ at all stationary points):

Global / Local Minimum (Reactants, Products, Reaction Intermediates): First derivatives $\partial E/\partial q_i = 0$ and **ALL Hessian second-derivative eigenvalues are positive ($\partial^2 E/\partial q_i^2 > 0$) → 0 imaginary frequencies**.

Transition State (First-Order Saddle Point): Energy is a **maximum along the reaction coordinate** and a minimum along all 3N − 7 perpendicular coordinates → **EXACTLY ONE imaginary frequency ($\nu_i < 0$ / negative force constant)**.

Ozone (O_3) PES Slide Case: Plots energy vs. O–O–O bond angle θ shows the **Global Minimum** at open bent ozone (**θ = 120.9°**, C_{2v}) and a higher-energy **Local Minimum** at cyclic ring **iso-ozone (θ = 60.0°**, D_{3h}) separated by a transition-state saddle point at θ ≈ 84°.

Water (H_2O) PES Slide Case: 3D surface over O–H bond length and H–O–H angle locates the global minimum at **r_{OH} = 0.958 Å and θ = 104.5°**, with a linear inversion barrier (θ = 180°) of ~11,000 cm^{-1}.



Visual Summary M5.1 (Slide Diagrams): *(Top Left) Homonuclear H_2 MO energy ladder (BO = 1.0, symmetric); (Top Right) Heteronuclear NO MO energy ladder showing lower-energy O orbitals, polarized MOs, and unpaired π^*_{2p} electron (BO = 2.5); (Bottom Left) Anharmonic diatomic PEC showing r_e, ZPE, D_o, and D_e; (Bottom Right) 3D Potential Energy Surface showing Reactant and Product minima connected via a Transition State saddle point.*

M5S4 · MEP COLOR CODE + 5-STEP PROTOCOL + SN2 DOUBLE-WELL

MEP Maps (NH_3), Geometry Optimization, Vibrational Modes & Gas vs. Aqueous SN2 Profiles

Molecular Electrostatic Potential (MEP) & NH_3 Map
Red → Green → Blue Scale

Definition: MEP V(r) maps the electrostatic interaction energy between a molecule's charge distribution (nuclei + electron density ρ) and a unit positive test charge (+1e) placed at point **r**, projected onto an electron-density isosurface (0.002 e Bohr^{-3}).

Universal Color Code (Increasing Potential): Red (Most Negative V < 0, electron-rich site → susceptible to Electrophilic attack / H-bond acceptor) < Orange < Yellow < **Green (Neutral / Non-polar V ≈ 0)** < **Blue (Most Positive V > 0, electron-deficient site → susceptible to Nucleophilic attack / H-bond donor)**.

Ammonia (NH_3) Slide Case: Intense **Red cap over the Nitrogen lone pair** (nucleophilic / basic protonation site) and **Blue regions over the three δ⁺ Hydrogen atoms**.

4-Step Geometry Optimization & Vibrational Analysis
H_2O (3), CO_2 (4), NH_3 (6), CH_4 (9)

4-Step Geometry Optimization Loop: (1) Input initial 3D coordinates (Z-matrix / Cartesian) + select method/basis set (e.g., B3LYP/6-31G(d)); (2) Solve SCF equations for energy E and analytical energy gradient (forces F = −∂E/∂R); (3) Check convergence criteria (Max Force, RMS Force, Max Displacement, RMS Displacement < threshold); (4) If not converged, update coordinates along the Newton–Raphson/BFGS step and repeat until forces vanish.

Vibrational Frequency Counts:

H_2O (Bent non-linear, N = 3): 3(3) − 6 = **3 modes** (symmetric stretch ~3657 cm^{-1}, asymmetric stretch ~3756 cm^{-1}, scissoring bend ~1595 cm^{-1}).

CO_2 (Linear, N = 3): 3(3) − 5 = **4 modes** (symmetric stretch 1388 cm^{-1} [IR-inactive], asymmetric stretch 2349 cm^{-1}, and doubly degenerate bend 667 cm^{-1} × 2).

NH_3 (Pyramidal non-linear, N = 4): 3(4) − 6 = **6 modes** | **CH_4 (Tetrahedral, N = 5):** 3(5) − 6 = **9 modes**.

5-Step Reaction Mechanism Protocol & S_N2 Profiles
Cl^- + CH_3Cl Gas vs. Solution

5-Step Computational Mechanism Protocol: (1) Optimize Reactants & Products (verify **0 imaginary frequencies**); (2) Scan reaction coordinate (relaxed PES scan) to locate candidate TS peak; (3) Perform Berny TS optimization and run Hessian frequency check to verify **exactly 1 imaginary frequency** along the bond-forming/breaking axis; (4) Run **Intrinsic Reaction Coordinate (IRC)** calculation in forward and reverse directions to confirm the TS connects the exact reactant and product valleys; (5) Compute zero-point and thermal corrections to obtain ΔH‡, ΔG‡, and rate constant *k* via Eyring equation.

Official Slide Case — S_N2 Reaction $Cl^- + CH_3Cl \rightarrow ClCH_3 + Cl^-$ (Gas Phase vs. Aqueous Solution):

Gas Phase (Double-Well Potential Profile): As bare Cl^- approaches CH_3Cl in vacuum, strong ion–dipole attraction first forms a stable pre-reaction **Ion–Dipole Complex $[Cl^-\cdots CH_3Cl]$** (energy minimum deeper than separated reactants!), then crosses the central **Pentacoordinate Trigonal-Bipyramidal Transition State $[Cl\cdots CH_3\cdots Cl]^-‡$**, and falls into a symmetric post-reaction ion–dipole complex before dissociating — producing a classic **Double-Well profile** with a low central barrier.

Polar Protic Solution (Unimodal Single-Barrier Profile): Water molecules strongly solvate the compact localized charge of free **Cl^-** via H-bonding much more than the charge-dispersed **$[Cl\cdots CH_3\cdots Cl]^-‡$** transition state. Because desolvating Cl^- costs substantial energy, the ion–dipole wells disappear and the reaction exhibits a **single high unimodal activation barrier ($\Delta G‡_{aq} \gg \Delta G‡_{gas}$)**, explaining why S_N2 reactions are 10^{12}–10^{15}× slower in water than in the gas phase!

M5S5 · 8 TARGET PROPERTIES + 6-STAGE AI PIPELINE

Contemporary Topics (Module 6): AI & Machine Learning in Semiconductor Discovery

Why Combine DFT with AI / Machine Learning?
10^6× Speedup

Traditional first-principles DFT/GW calculations of crystal bandgaps and mobilities take hours to days per compound, limiting screening to a few hundred structures. Training **Graph Neural Networks (GNNs like CGCNN, MEGNet, ALIGNN)** and ensemble ML models on quantum databases (**Materials Project, OQMD, AFLOW, NOMAD**) predicts properties of millions of candidate crystals in **milliseconds**.

8 Target Semiconductor Properties Predicted by AI: (1) Electronic Bandgap (**E_g**, direct vs. indirect); (2) Thermodynamic stability (**Energy above convex hull E_{hull}** & formation enthalpy ΔH_f); (3) Electron & hole effective masses (m^*_e, m^*_h) and carrier mobilities; (4) Dielectric constant (ε_r) & breakdown field; (5) Optical absorption coefficient $\alpha(\omega)$; (6) Dopability & defect tolerance; (7) Lattice thermal conductivity (κ_L); (8) Mechanical elastic moduli.

6-Stage AI High-Throughput Discovery Workflow

Forward Screening & Inverse Design

Stage 1 (Data Curation): Harvest DFT-computed crystal structures and properties from Materials Project / OQMD (>150,000 compounds).

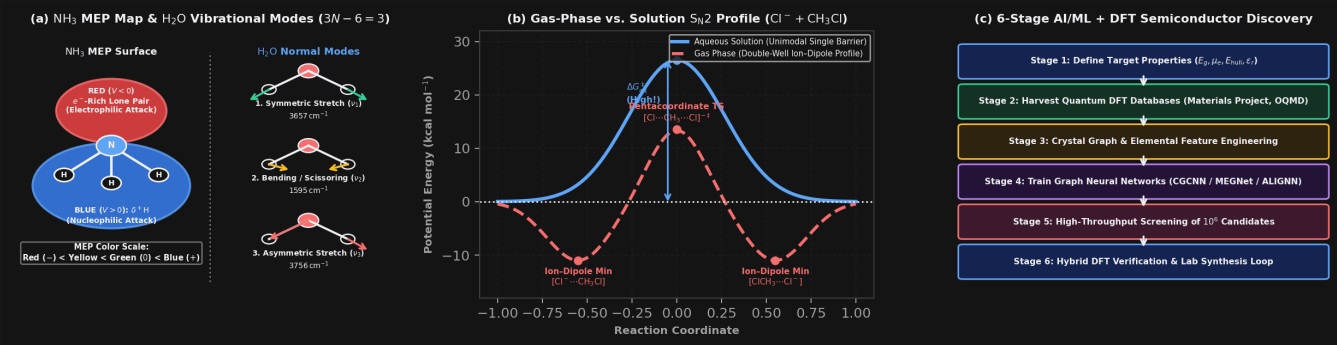
Stage 2 (Featurization / Descriptors): Convert crystal structures into machine-readable chemical composition vectors (Magpie elemental features) and 3D crystal graphs (atoms = nodes, bonds = edges).

Stage 3 (ML Model Training & Validation): Train GNNs / Random Forests / Generative VAEs-Diffusion models using cross-validation.

Stage 4 (High-Throughput Screening / Inverse Design): Filter 10^6 hypothetical compositions down to top 10–50 candidates matching target criteria (e.g., E_{hull} < 25 meV/atom, direct E_g = 1.3–1.5 eV, non-toxic earth-abundant elements).

Stage 5 (DFT Verification): Validate top ML hits with high-level hybrid DFT (HSE06) and phonon stability calculations.

Stage 6 (Robotic Synthesis & Active Learning Loop): Synthesize and test top hits experimentally, feeding results back into the training set.



Visual Summary M5.2 (Slide Diagrams): *(Left) Molecular Electrostatic Potential (MEP) map of NH_3 showing the negative red region over the N lone pair and positive blue regions over H atoms; (Center) S_N2 reaction profile of Cl^- + CH_3Cl comparing the gas-phase double-well curve against the aqueous solution unimodal curve; (Right) 6-stage AI/ML + DFT high-throughput workflow for semiconductor material discovery.*

M5S6 · 100% SOLVED MULTI-PART QUESTIONS

Complete Official VIT SAS Module 5 Model Question Bank Solutions (Q1–Q7)

Official Model Q1 & Q2 — Scope, Merits, Electronic Structure Methods & Basis Sets

VIT SAS Question Bank · 10 Marks

Problem: (Q1) Define computational chemistry, classify its major branches (QM vs. MM/MD), state Born–Oppenheimer approximation, and list 7 merits over experiment. (Q2) Compare HF, DFT, MP2, CI, CCSD(T), and CASSCF, and explain Minimal (STO-3G), Split-Valence (6-31G), Polarized (*), Diffuse (+), and ECP (LANL2DZ) basis sets.

Complete Exam Blueprint: Directly reproduce the structured tables and derivations in **Section M5S1**: (1) TISE $\hat{H}\Psi = E\Psi$ with Born–Oppenheimer decoupling ($M_{nuc} \geq 1836\ m_e$, $\hat{T}_n = 0 \rightarrow E(R)$); (2) 7-row Electronic Structure Methods table highlighting HF mean-field (no dynamic correlation), DFT $\rho(r)$ (~N^3 workhorse), MP2 (~N^5), CCSD(T) (~N^7 gold standard), and CASSCF (multi-reference bond breaking); (3) 6-row Basis Set table explaining how **6-31+G(d,p)** combines a 6-PGTO core, 3/1 split valence, *d/p* polarization for angular flexibility, and + diffuse functions for anions/lone pairs.

Official Model Q3 — Homonuclear (H_2) vs. Heteronuclear (NO) MO Diagrams & LCAO Principles

VIT SAS Question Bank · 10 Marks

Problem: State the 3 LCAO conditions. Construct and compare the Molecular Orbital diagrams of H_2 and NO, explaining why oxygen orbitals lie lower than nitrogen orbitals in NO, and calculate their bond orders and magnetic behavior.

Complete Exam Blueprint (Section M5S2 + Visual M5.1):

3 LCAO Criteria: Comparable energy, strong spatial overlap, identical symmetry along the internuclear axis.

H_2 (2 e^-): Symmetric 1s combination → $(\sigma_{1s})^2(\sigma^*_{1s})^0$; **BO = ½(2−0) = 1.0**; diamagnetic; HOMO = σ_{1s}, LUMO = σ^*_{1s}.

NO (11 valence e^-): Because χ_O (3.44) > χ_N (3.04), O 2s/2p AOs lie lower in energy than N 2s/2p AOs. Bonding MOs are polarized toward O ($c_O > c_N$) and antibonding MOs toward N ($c_N > c_O$). Valence configuration:

$(\sigma_2s)^2(\sigma^*_2s)^2(\pi_2px)^2(\pi_2py)^2(\sigma_2pz)^2(\pi^*_2px)^1$. **BO = ½(8−3) = 2.5**; **paramagnetic radical** with 1 unpaired electron in π^*_{2p}.

Official Model Q4 & Q5 — PEC (H_2, HCl) vs. PES (Ozone, H_2O), Minima vs. Saddle Points & Vibrational Modes

VIT SAS Question Bank · 10 Marks

Problem: (a) Differentiate between a 1D Potential Energy Curve (PEC) and a multi-dimensional Potential Energy Surface (PES) / Hypersurface. (b) Explain the PEC of H_2/HCl and the PES of Ozone (O_3) and Water (H_2O). (c) Calculate the vibrational degrees of freedom for H_2O, CO_2, NH_3, CH_4, and C_6H_6, and state how Hessian frequency analysis distinguishes a minimum from a transition state.

Complete Exam Blueprint (Section M5S3 & M5S4):

PEC vs. PES: Diatomics have 1 internal coordinate (r) → 2D curve; polyatomics have **3N−6** (non-linear) or **3N−5** (linear) internal coordinates → 3D surface (for 2 coords) or **Hypersurface** (for ≥3 coords).

Vibrational Mode Counts: H_2O (non-linear, N=3): **3 modes**; CO_2 (linear, N=3): **4 modes**; NH_3 (non-linear, N=4): **6 modes**; CH_4 (non-linear, N=5): **9 modes**; Benzene C_6H_6 (non-linear, N=12): 3(12)−6 = **30 modes**.

Hessian Frequency Rule: Both minima and transition states have zero first derivative (∇E = 0). A **Local/Global Minimum (Reactant/Intermediate/Product) has 0 imaginary frequencies** (all positive force constants), whereas a **Transition State (1st-order saddle point) has EXACTLY 1 imaginary frequency** along the reaction coordinate.

Official Model Q6 & Q7 — NH_3 MEP Map, Gas vs. Aqueous S_N2 Profile & AI Semiconductor Discovery

VIT SAS Question Bank · 10 Marks

Problem: (a) Explain the Molecular Electrostatic Potential (MEP) surface of NH_3 and its color scale. (b) Compare the potential energy profile of the bimolecular nucleophilic substitution reaction **$Cl^- + CH_3Cl \rightarrow ClCH_3 + Cl^-$** in the gas phase versus aqueous solution. (c) Describe the 6-stage AI/ML + DFT workflow for discovering new semiconductors.

Complete Exam Blueprint (Section M5S4, M5S5 & Visual M5.2):

NH_3 MEP: Red (negative V < 0) over the N lone-pair cap (electrophilic attack / protonation site) and Blue (positive V > 0) over the three H atoms (nucleophilic attack / H-bond donor).

Gas vs. Solution S_N2 Profile: Gas phase gives a **Double-Well profile** because unsolvated Cl^- forms stable pre- and post-reaction ion–dipole complexes **$[Cl^-\cdots CH_3Cl]$** on either side of the trigonal-bipyramidal TS **$[Cl\cdots CH_3\cdots Cl]^-‡$**. In aqueous solution, strong H-bond solvation stabilizes compact Cl^- much more than the charge-dispersed TS, wiping out the ion–dipole wells and producing a **single high unimodal barrier ($\Delta G‡_{aq} \gg \Delta G‡_{gas}$)**.

AI/ML Semiconductor Workflow: Data curation (Materials Project/OQMD) → Crystal graph featurization → GNN/ML training → High-throughput screening of E_g, E_{hull}, and μ → DFT verification → Experimental synthesis & active learning.